

Shengwei Tian

Google Scholar 

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EDUCATION

Harbin Institute of Technology (HIT)

Mar 2026 - Mar 2030 (*expected*)

Doctor of Philosophy, Computer Science and Technology (Supervisor: Xiaopeng Fan)

Research Topics: Multimodal Representation Learning, Computer vision, AI4S

University of Manchester (UoM)

Sep 2023 - Dec 2024

MSc Data Science

Related courses: Statistics and Machine Learning, Text Mining, Computer vision.

China Jiliang University (CJLU)

Sep 2019 - Jul 2023

Bachelor of Engineering in Electronic Information Science and Technology

RESEARCH INTERESTS

My research interests focus on developing robust and generalizable AI models for understanding complex multimodal and structured data. I am particularly interested in foundation models and multimodal representation learning, with an emphasis on model robustness, reliability, and intellectual property protection. I am also interested in applying advanced deep learning methods to scientific problems, especially single-cell and biological data, to uncover meaningful representations and cross-domain patterns from large-scale datasets.

RESEARCH EXPERIENCE

Institute of Systems Medicine, Chinese Academy of Medical Sciences / Suzhou Institute of Systems Medicine (ISM)

Jul 2025 - Mar 2026

Supervisors: Prof. Dongshen Chen

Research Assistant

- Contributed substantially to multiple computational single-cell research projects under the supervision of Prof. Chen, taking part in analytical workflow design, large-scale data processing, cell-type annotation, downstream analysis, biological interpretation, and research presentation.
- Played a role in two ongoing research projects currently being prepared for journal submission, independently driving key components of the computational analyses, troubleshooting analytical challenges, interpreting experimental results, and contributing to the development of research findings and manuscript preparation.
- Contributed to *LACA: a cross-species single-cell resource for exploring longevity and aging associated cellular and molecular programs*, a large-scale cross-species single-cell study investigating conserved and species-specific cellular and molecular programs associated with longevity and aging, accepted as a co-first author by Nucleic Acids Research (NAR) in 2026.

Zhejiang University Binjiang Institute

Jan 2025 - Jun 2025

Supervisors: Prof. Meng Han

Research Assistant

- Conducted research on intellectual property protection for large language models, focusing on robust and stealthy model fingerprinting under black-box access and adversarial model modifications.
- Designed and conducted experiments across multiple LLM architectures to evaluate fingerprint effectiveness, robustness, stealthiness, and false-positive behavior, and benchmarked CTCC against existing fingerprinting approaches under diverse attack and modification settings.
- Contributed to the resulting paper, "CTCC: A Robust and Stealthy Fingerprinting Framework for Large Language Models via Cross-Turn Contextual Correlation Backdoor," published at EMNLP 2025 (Main Conference).

PUBLICATIONS

1. “LACA: A Cross-Species Single-Cell Resource for Exploring Longevity and Aging Associated Cellular and Molecular Programs.” *Nucleic Acids Research (NAR)*, 2026. **Accepted**. (Co-first author.)
2. Zhenhua Xu, Xixiang Zhao, Xubin Yue, **Shengwei Tian**, Changting Lin, and Meng Han. “CTCC: A Robust and Stealthy Fingerprinting Framework for Large Language Models via Cross-Turn Contextual Correlation Backdoor.” *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*. **Accepted**.
3. **Shengwei Tian**, Wenrui Li, Rui Zhao, Xiaopeng Fan. “scRetrieval: Semantic-Aligned Cross-Modal Representation Learning for Single-Cell Transcriptome–Text Retrieval.” To be submitted to *Nature Machine Intelligence (NMI)*.
4. Yeyu Chai, Wenrui Li, Rui Zhao, **Shengwei Tian**, Hengyu Man, and Xiaopeng Fan. “HEROS: Hierarchical Semantic Forest with Hyperbolic Multi-View Saliency-Aware Aggregation Network for Text–3D Heritage Retrieval.” *The 41st AAAI Conference on Artificial Intelligence (AAAI 2027)*, 2027. **Under Review**.
5. Zitong Hu, Wenrui Li, **Shengwei Tian**, Kui Jiang, Boyun Li, Jiantao Zhou, and Xiaopeng Fan. “PFE-Net: A Dual-Branch Physics-Constrained Frequency Enhancement Network for Underwater Image Enhancement.” *The 41st AAAI Conference on Artificial Intelligence (AAAI 2027)*, 2027. **Under Review**.
6. Zhe Yang, Wenrui Li, **Shengwei Tian**, Hongtao Chen, Jiantao Zhou, and Xiaopeng Fan. “Hierarchical Semantic-Constrained Heterogeneous Graph for OV-AVEL.” *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*. **Under Review**.

SELECTED AWARDS

Outstanding Student Scholarship, China Jiliang University	2021–2022
First Prize, Financial Innovation Competition, China Jiliang University	2020
Second Prize, Undergraduate Securities Investment Competition, China Jiliang University	2020
Volunteer Project Leader, Intercultural and Innovation Management Programme, Manchester, UK	2024
Volunteer Project Leader, Chunhui Cup Chinese Overseas Students Innovation and Entrepreneurship Competition, Manchester, UK	2024